

External assessment 2022

Multiple choice question book

Chemistry

Paper 1

General instruction

- Work in this book will not be marked.



Queensland
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Section 1

QUESTION 1

Identify the type of reaction that occurs when ethene undergoes polymerisation to form polyethene.

- (A) addition
- (B) elimination
- (C) substitution
- (D) condensation

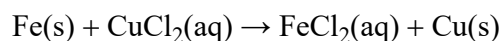
QUESTION 2

Structural isomers are compounds with the same molecular formula but a different

- (A) molar mass.
- (B) molecular mass.
- (C) empirical formula.
- (D) arrangement of atoms.

QUESTION 3

Which option is true for the redox equation?



- (A) Fe is oxidised and Cu is the oxidising agent
- (B) Fe is oxidised and Cu^{2+} is the oxidising agent
- (C) Fe^{2+} is oxidised and Cu is the oxidising agent
- (D) Fe^{2+} is oxidised and Cu^{2+} is the oxidising agent

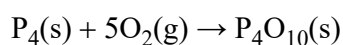
QUESTION 4

Which pair of reagents would react to form a glycosidic bond?

- (A) lysine and aniline
- (B) glucose and galactose
- (C) methanol and butanoic acid
- (D) glycerol and sodium hydroxide

QUESTION 5

Phosphorus pentoxide is prepared by burning phosphorus in oxygen.

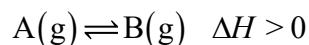


Calculate the percentage yield if 10.0 g of P_4O_{10} is produced when 0.200 mol of P_4 and 0.200 mol of O_2 are reacted.

- (A) 2.0%
- (B) 3.5%
- (C) 17.6%
- (D) 88.0%

QUESTION 6

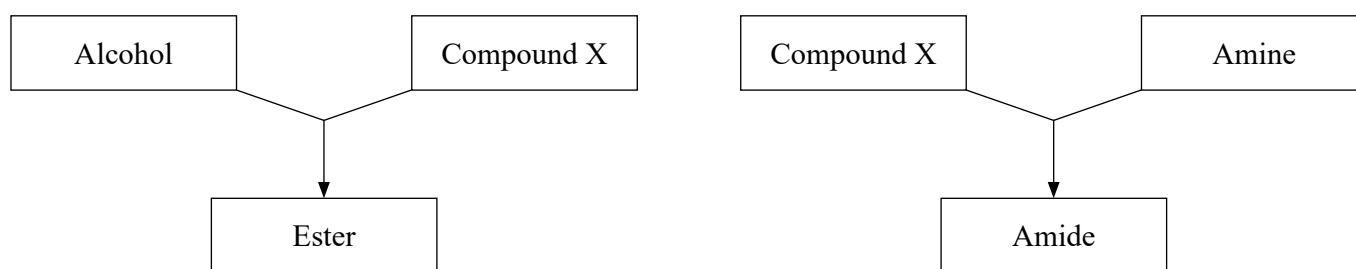
The equilibrium concentration of A is 2.8×10^{-4} M and B is 1.2×10^{-4} M.



Which option represents the ratio of molecules present in a sample of the gaseous mixture when the temperature is decreased and a new equilibrium established?

- (A) 8 molecules of A and 2 molecules of B
- (B) 5 molecules of A and 5 molecules of B
- (C) 3 molecules of A and 7 molecules of B
- (D) 2 molecules of A and 8 molecules of B

QUESTION 7

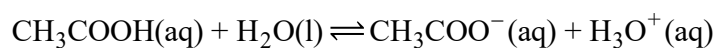


Compound X in these reaction pathways is

- (A) a ketone.
- (B) an alkene.
- (C) an aldehyde.
- (D) a carboxylic acid.

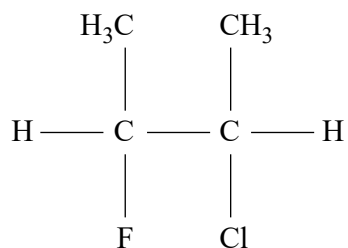
QUESTION 8

Predict how the system shown will respond when a small amount of aqueous sodium hydroxide is added.



- (A) Equilibrium shifts to the left and the pH decreases.
- (B) Equilibrium shifts to the right and the pH increases.
- (C) Equilibrium shifts to the left and the pH remains the same.
- (D) Equilibrium shifts to the right and the pH remains the same.

QUESTION 9



The IUPAC name for this molecule is

- (A) 2-chloro-3-fluorobutane.
- (B) 2-fluoro-3-chlorobutane.
- (C) 2-dimethyl-1-chloro-2-fluoroethane.
- (D) 1,2-dimethyl-1-fluoro-2-chloroethane.

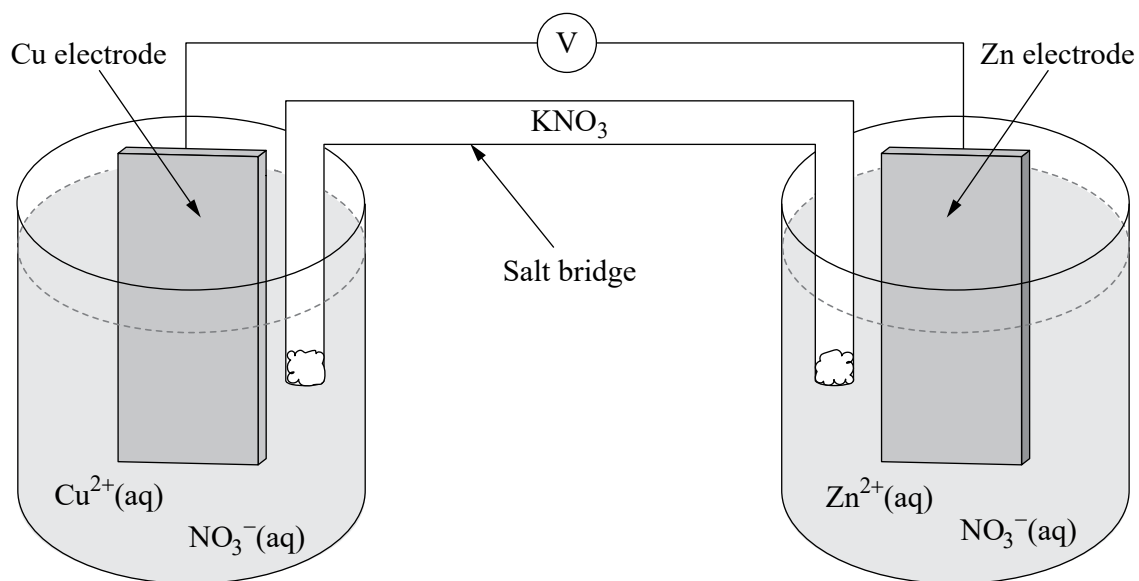
QUESTION 10

The midpoint of the colour change of a weak acid indicator occurs when

- (A) $[\text{In}^-] = [\text{H}^+]$
- (B) $[\text{In}^-] = [\text{HIn}]$
- (C) $[\text{H}^+] = [\text{OH}^-]$
- (D) $[\text{HIn}] = [\text{OH}^-]$

QUESTIONS 11–12

These questions refer to the diagram shown.



QUESTION 11

Determine the species that travels through the salt bridge towards the reduction half-cell in the electrochemical cell at standard conditions.

- (A) zinc ions
- (B) nitrate ions
- (C) copper ions
- (D) potassium ions

QUESTION 12

The zinc electrode

- (A) gains electrons and acts as the anode.
- (B) acts as the cathode and has a positive charge.
- (C) undergoes reduction and has a negative charge.
- (D) is oxidised and donates electrons to the copper ions.

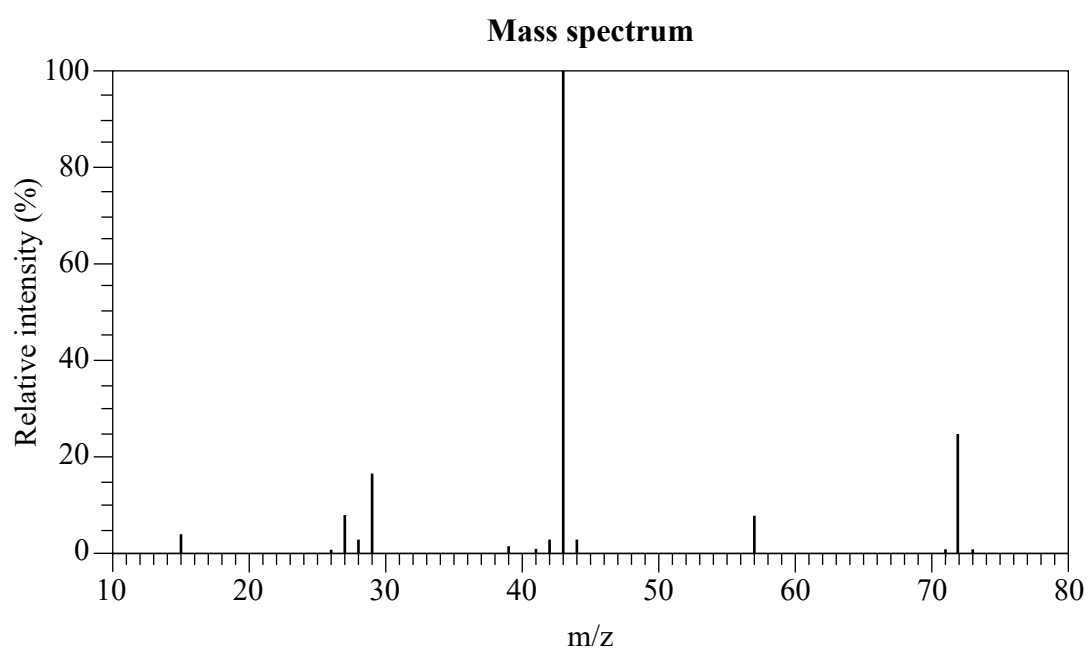
QUESTION 13

Determine the K_a of an unknown weak acid (HA) with an aqueous concentration of 0.12 M and a pH of 3.2.

- (A) 5.2×10^{-3}
- (B) 6.3×10^{-4}
- (C) 3.3×10^{-6}
- (D) 4.0×10^{-7}

QUESTION 14

The mass spectrum for Compound X is found to have signals at the following m/z values.

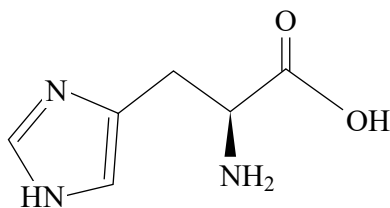


Compound X is

- (A) butanal.
- (B) butanol.
- (C) butanone.
- (D) butanoic acid.

QUESTION 15

The structure of an amino acid is shown.



This molecule contains an amine group and a

- (A) carboxyl group.
- (B) hydroxy group.
- (C) methyl group.
- (D) ketone group.

QUESTION 16

Determine the oxidation state of manganese in MnO_4^- .

- (A) +1
- (B) +2
- (C) +7
- (D) +8

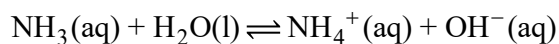
QUESTION 17

Identify the redox reaction.

- (A) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (B) $\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca}(\text{OH})_2(\text{s})$
- (C) $\text{Cl}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{HCl}(\text{aq}) + \text{HClO}(\text{aq})$
- (D) $\text{NaOH}(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

QUESTION 18

Determine the K_b expression for the weak base shown in the equilibrium equation.



(A) $K_b = \frac{[\text{NH}_3][\text{H}_2\text{O}]}{[\text{NH}_4^+]}$

(B) $K_b = \frac{[\text{NH}_3][\text{H}_2\text{O}]}{[\text{OH}^-]}$

(C) $K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$

(D) $K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$

QUESTION 19

Three voltaic cells are constructed with metal Q as one electrode and metals R, S or T as the other electrode. The potential differences for the cells are shown in the table.

Voltaic cell	Half-cell	Half-cell	Potential difference (V)
1	$\text{Q}(\text{s}) / \text{Q}^{2+}(\text{aq})$	$\text{R}^+(\text{aq}) / \text{R}(\text{s})$	1.18
2	$\text{Q}(\text{s}) / \text{Q}^{2+}(\text{aq})$	$\text{S}^{2+}(\text{aq}) / \text{S}(\text{s})$	0.72
3	$\text{T}(\text{s}) / \text{T}^{3+}(\text{aq})$	$\text{Q}^{2+}(\text{aq}) / \text{Q}(\text{s})$	0.95

The relative strength of the reducing agents from strongest to weakest is

(A) $\text{T} > \text{Q} > \text{S} > \text{R}$

(B) $\text{S} > \text{Q} > \text{T} > \text{R}$

(C) $\text{R} > \text{Q} > \text{S} > \text{T}$

(D) $\text{Q} > \text{R} > \text{T} > \text{S}$

QUESTION 20

1	$\left(\text{CH}_2 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} - \text{CH}_2 - \overset{\text{CH}_3}{\overset{ }{\text{CH}}} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} - \text{CH}_2 - \overset{\text{CH}_3}{\overset{ }{\text{CH}}} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} \right)$
2	$\left(\text{CH}_2 - \overset{\text{CH}_3}{\overset{ }{\text{CH}}} - \text{CH}_2 - \overset{\text{CH}_3}{\overset{ }{\text{CH}}} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} - \text{CH}_2 - \overset{\text{CH}_3}{\overset{ }{\text{CH}}} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{ }{\text{CH}}} \right)$

The two forms of polypropene shown are

	1	2
(A)	syntactic	atactic
(B)	isotactic	atactic
(C)	isotactic	syntactic
(D)	atactic	syntactic

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